

Year 8 Science – Booklet 23: Biology

Photosynthesis — ANSWERS

Note: this booklet's questions are identical to Year 8 Booklets 14 and 21 (Photosynthesis), so the same worked answers apply throughout.

Quick Test (p.27)

1. Glucose and oxygen.
2. In the leaves (mainly in the chloroplasts of the palisade cell layer).
3. Nitrates, phosphates and potassium.

Section A – Multiple Choice (5 marks)

1. a) during the day
2. c) xylem
3. d) palisade cells
4. d) nitrate
5. d) stunted growth

Section B (7 marks)

1. Match the mineral to its deficiency symptom (3 marks)

Nitrate → poor root growth and purple young leaves

Phosphate → yellow leaves with dead spots

Potassium → stunted growth and yellow older leaves

2. Labelled leaf section (4 marks)

- a). Palisade cells (contain the most chloroplasts).
- b). Stoma (the opening that lets carbon dioxide into the leaf).
- c). The waterproof layer (cuticle).
- d). The leaf vein (contains the xylem and phloem tubes).

Section C (7 marks)

1. Pondweed photosynthesis experiment

- a). Oxygen.
- b). Collect the gas in a test tube and test it with a glowing splint – if it relights, this confirms the gas is oxygen.
- c). Light and carbon dioxide (2 marks).
- d). Glucose.

2. Iodine test on pondweed leaves

- a). Blue-black.
- b). Starch – showing that photosynthesis had taken place, since the glucose made is stored as starch.